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A Monte Carlo estimation of tissue optical properties for use in laser dosimetry

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Abstract. In certain clinical situations, such as photodynamic therapy, light dosimetry should be considered. The propagation of light in tissues is influenced by fundamental or microscopic optical properties, namely absorption μ_a and scattering μ_s coefficients, refractive index n and anistropy factor g. These optical parameters can be determined experimentally by direct and/or indirect methods when tissue macroscopic properties, such as reflectance, transmittance or collimated transmittance from a tissue slab, are measured. The method described in this work provides graphical, and in simple cases analytical, 'inverse' solutions to determine tissue microscopic properties from measured macroscopic parameters. The graphs necessary for this inversion have been calculated and are provided. The method can be applied in either direct or indirect techniques and it does not depend on limitations introduced by assumptions and approximations when using theoretical models. It can also be applied for any tissue type, detector geometry and experimental apparatus. The accuracy of the method is very good over a wide range, unlimited in practice, of values of optical properties. Finally, the results of this work are in good agreement with theoretical and experimental results of other investigators.

1. Introduction

The propagation of light in tissues is of great interest in many areas of medicine and biology, as the number of applications of light, and in particular of laser beams, in such areas is growing. In the field of photodynamic therapy (PDT), for example, it is important to know the distribution of light within tissues. Also in the area of fluorescence imaging and transmission imaging, the transmittance and reflectance of a tissue slab must be known.

The propagation of light in tissues is influenced by the microscopic optical properties of tissues, namely the absorption and scattering coefficient, the anisotropy factor and the refractive index. Typically, these can be determined by expressing the optical properties in terms of measurable macroscopic quantities, such as diffuse transmittance, diffuse reflectance and collimated transmittance, using diffusion theory (Patterson *et al* 1991) or applying an adequate computational method such as the adding–doubling method (Prahl *et al* 1993, van Hillegersberg *et al* 1993, Pickering *et al* 1993a) or the Monte Carlo method (Key *et al* 1991, Peters *et al* 1990).

The macroscopic properties are measured *in vivo* and/or *in vitro* by direct and/or indirect techniques (Cheong *et al* 1990, Wilson *et al* 1987). Direct techniques are based on measurements in tissue samples thin enough that single-photon scattering occurs. The microscopic coefficients are obtained directly from the fraction of light absorbed or scattered by the sample. In this case, the results do not depend on the model of light propagation, but on the other hand sample preparation is difficult since tissue thicknesses must be less

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than the mean free path length, which is a few micrometres (Flock *et al* 1987, Wilson *et al* 1987, Peters *et al* 1990). Indirect techniques are based on measurements in bulk tissues of diffuse transmittance and reflectance. These two parameters, together with the sample thickness, are combined with an appropriate light propagation model to determine the microscopic coefficients. Although sample preparation is simple, since tissues do not have to be optically thin, the results depend on the model of light propagation, which may introduce errors into calculations (Wilson *et al* 1987). Most experiments using indirect techniques have produced very diverse results, which will be referred to in the discussion and which show the shortcomings of the method (Peters *et al* 1990, Flock *et al* 1989, Pickering *et al* 1993a, b, Driver *et al* 1991, van Hillegersberg *et al* 1993).

In this paper we devise a Monte Carlo code to compute tissue optical microscopic coefficients, when macroscopic properties have been measured. We also propose a practical method based on our Monte Carlo results by which one can determine microscopic coefficients using data from macroscopic measurements. The method provides graphical, and in some instances analytical, 'inverse' solutions, deriving the microscopic properties from measurements of the macroscopic parameters. Thus, the results of this work can be used as a reference for microscopic property determination following optical parameter measurements using indirect techniques, in any type of tissue.

2. Materials and method

A Monte Carlo code was written in FORTRAN computer language and run on a CYBER 170-730 computer. The code simulated the irradiation of a three-dimensional parallel slab of tissue of a given thickness with a collimated light beam. This is the usual experimental arrangement for measuring the tissue optical parameters with indirect techniques.

In this work the variable-step-size Monte Carlo method was chosen (Prahl 1988). Photons were generated via a random number algorithm and traced through the tissue volume using a variable step size. The step size S was chosen in such a way that the photon is forced either to scatter or be absorbed, according to formulae $S_{\mathrm{abs}} = -(\ln R)/\mu_{\mathrm{a}}$ and $S_{\mathrm{scatt}} = -(\ln R)/\mu_{\mathrm{s}}$, where R is a random number $[0, 1]$. If a photon was scattered then a new photon direction was chosen, based on the Henyey–Greenstein phase function. Otherwise it was absorbed and a new photon was generated. When the photon crossed the front or the rear boundary of the sample, a new photon was generated.

When the sample consisted of several layers, for example a tissue slab sandwiched between two glass slides, then the photon either crossed the boundary according to Snell's law, or was internally reflected according to Fresnel's law, depending on the reflection probability. In both cases the position and direction of the photon were adjusted accordingly. This Monte Carlo method needs a large number of photon histories in order to give accurate results and consequently it needs more computation time, which was not a critical parameter in this work. The accuracy of the method has been proved to be excellent as will be shown.

Many investigators in this field are using variance reduction techniques (Wilson and Adam 1983, Keijzer *et al* 1989, Key *et al* 1991, Peters *et al* 1990) or hybrid models (Wang and Jacques 1993) in order to reduce the computation time and to achieve better statistics. These techniques assign a weight to each photon as it enters the sample, which is reduced by the probability of absorption after each propagation step. The present code, as was described, was chosen in order to be used in multilayer problems when necessary, since the variance reduction techniques are awkward to implement in such cases (Prahl 1988).

In this work, for each run 100 000 photons were generated. The Cartesian coordinate system that was used had its origin at the point where the central ray of the light beam intersects the slab. The tissue slab was assumed to be exposed to the light beam without being compressed with transparent plates. The time needed for each run depended on the input data (optical coefficients, slab thickness etc) and generally was a few minutes. Since the results are not intended for immediate clinical use, but for the determination of microscopic parameters to be used in clinical laser dosimetry, the computation time was not considered to be critical.

The input entry included the beam shape, which could be collimated, diverging, focused beam or point source, the beam diameter, the location of the entrance point, the incident angle and in the case of collimated transmittance calculations the beam cross-sectional area at the exit. Also, the tissue slab thickness and its microscopic optical properties, that is, the absorption and scattering coefficients, the anisotropy factor g and the refractive index, were included at the input data of the code.

In this work, a collimated light beam with uniform irradiance, 1 cm in diameter, was simulated. The beam was incident normally to the slab, the surface of which was wide enough, compared to the beam's cross-section, that it could be considered as infinite.

In order to simulate a wide range of tissue types, the scattering coefficient μ_s was varied from 0.5 cm^{-1} to 20 cm^{-1}, the absorption coefficient μ_a from 0.0156 cm^{-1} to 4 cm^{-1} and the anisotropy factor g from 0.8 to unity. Therefore, the reduced scattering coefficient μ_s^* ($\mu_s^* = \mu_s[1-g]$) and albedo a ($a = \mu_s/[\mu_s + \mu_a]$) were within the ranges of 0–4 cm^{-1} and 0.021–39.22, respectively.

Most tissue types have microscopic optical coefficients varying in the above mentioned ranges (Cheong *et al* 1990).

To take into consideration internal reflection and refraction at air–tissue and tissue–air boundaries the tissue refractive index was set equal to 1.4. This value was chosen since the refractive index for most tissue types is in the range of 1.36–1.42, with the exception of adipose tissue, which has $n = 1.45$ (Bolin *et al* 1989). In order to check the difference introduced by refractive index variations, two different values for n ($n = 1.36$ and $n = 1.45$) were used as an input data to the code. Although the difference found for specular reflectance was up to 15%, the differences for the diffuse transmittance and total reflectance were less than 4% in both cases.

Finally, the slab thickness was 1 cm. This thickness was chosen to show that even in an unfavourable situation of a thick sample, the proposed method gives dependable results.

3. Results

For each set of input data (μ_s, μ_a, g and slab thickness *of 1.0 cm*) the following quantities were calculated: diffuse transmittance T; Fresnel or specular reflectance f; diffuse reflectance γ; total reflectance R; absorption A and collimated transmittance T_c.

In the present work the above quantities have the following meaning. T is taken as the ratio of photons transmitted through the slab and the photons incident on it, f as the portion of photons specularly reflected at the air–tissue interface, γ as the portion of the incident photons escaping from the upper side of the slab after having entered the slab, R as the sum of γ and f, A as the portion of the incident photons that were absorbed inside the slab and T_c as the ratio of photons escaping from the slab inside the beam diameter and the photons incident on the slab. All these macroscopic quantities are shown in a sketch in figure 1.

The above defined quantities must satisfy the relation

$$T + f + \gamma + A = 1$$

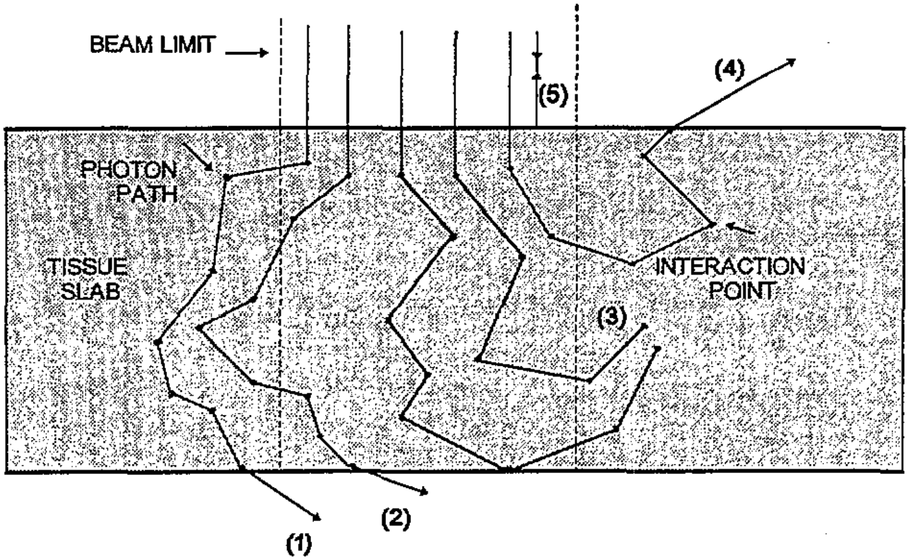

Figure 1. Types of photon path passing through a tissue slab. The paths considered in the Monte Carlo code are the following: diffuse transmittance T (1 + 2); collimated transmittance T_c (2); fraction absorbed A (3); diffuse reflectance γ (4); specular (Fresnel) reflectance f (5); total reflectance R (4 + 5).

which was also used to check the correctness of the algorithm.

In order to examine the influence of μ_a, μ_s and g for a given slab thickness on the above mentioned macroscopic quantities, the Monte Carlo code was run several times, each time for a different 'triplet' $[\mu_a, \mu_s, g]$ of coefficients. Absorption coefficient μ_a, scattering coefficient μ_s and anisotropy factor g varied in a sequence within the ranges mentioned in the previous section. For each run all the above macroscopic quantities were calculated. The reproducibility of the code was checked by running the program several times. For 100 000 histories in each run, the overall reproducibility of all calculated quantities was found to be better than 99%.

The results of the calculations for the diffuse transmittance T and total reflectance R versus reduced scattering coefficient μ_s^* for absorption coefficient $\mu_a = 0.25$ cm^{-1} are presented in figure 2. The reduced scattering coefficient μ_s^* was chosen as an independent variable in order to reduce the number of independent variables, as it was found that, in the range that g was varied, this similarity transform in scattering was applied without affecting the values of T and R. Figure 2 also shows the plots fitted to the data. For this fitting use was made of formulae

$$T = T_1/(\mu_s^* + T_2)^{T_3} \tag{1}$$

$$R = R_1 + R_2 {\mu_s^*}^{R_3} \tag{2}$$

where T_1, T_2, T_3, R_1, R_2 and R_3 are constants depending on the particular values of μ_a and slab thickness that were used.

In the same manner, using different values of μ_a, regression analysis provides values for the above constants, which are presented in tables 1 and 2. R^2 was always better than 99%.

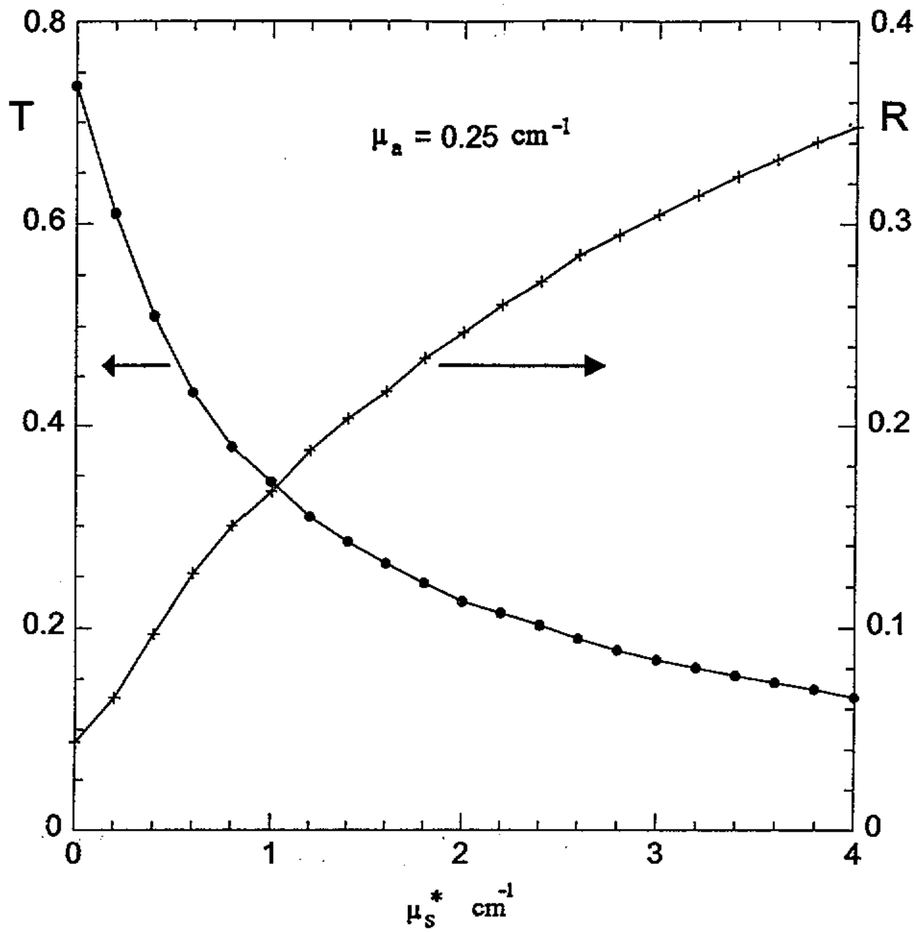

Figure 2. Diffuse transmittance and total reflectance as a function of μ_s^* for $\mu_a = 0.25$ cm^{-1} and a slab thickness of 1.0 cm. The curves have been fitted to the Monte Carlo results, which are shown as points (●, diffuse transmittance, T, and ×, total reflectance, R).

Table 1. The fitting of diffuse transmittance to the model $T = T_1/(\mu_s^* + T_2)^{T_3}$ for a slab thickness of 1.0 cm, and for μ_s^* with 0–4 cm^{-1}.

μ_a (cm^{-1})	T_1	T_2	T_3
0.0156	0.77 ± 0.01	0.70 ± 0.02	0.514 ± 0.008
0.0312	0.73 ± 0.01	0.65 ± 0.02	0.531 ± 0.009
0.0615	0.68 ± 0.01	0.63 ± 0.02	0.57 ± 0.01
0.125	0.64 ± 0.01	0.68 ± 0.02	0.71 ± 0.01
0.25	0.62 ± 0.02	0.84 ± 0.03	0.97 ± 0.02
0.5	0.71 ± 0.05	1.15 ± 0.05	1.47 ± 0.04
1	1.4 ± 0.3	1.8 ± 0.1	2.4 ± 0.1
2	2.2 ± 0.6	2.3 ± 0.2	3.5 ± 0.3
4	5.0 ± 0.8	3.1 ± 0.3	5.2 ± 0.4

Table 2. The fitting of total reflectance to the model $R = R_1 + R_2\mu_s^{*R_3}$ for a slab thickness of 1.0 cm, and for μ_s^* within 0–4 cm^{-1}.

μ_a (cm^{-1})	R_1	R_2	R_3
0.0156	0.03 ± 0.02	0.31 ± 0.02	0.45 ± 0.03
0.0312	0.03 ± 0.02	0.29 ± 0.02	0.46 ± 0.03
0.0615	0.03 ± 0.01	0.25 ± 0.02	0.49 ± 0.03
0.125	0.03 ± 0.01	0.20 ± 0.01	0.52 ± 0.03
0.25	0.032 ± 0.006	0.136 ± 0.006	0.63 ± 0.03
0.5	0.030 ± 0.003	0.077 ± 0.003	0.76 ± 0.03
1	0.028 ± 0.001	0.038 ± 0.001	0.93 ± 0.02
2	0.027 ± 0.001	0.016 ± 0.001	1.09 ± 0.02
4	0.027 ± 0.001	$0.006 \pm < 0.001$	1.22 ± 0.03

Using the constants from the above tables 1 and 2, two families of graphs were calculated, one for T and a second for R, versus μ_a and μ_s^*. These graphs are presented in the three-dimensional plots of figure 3(a) and (b), respectively. As can be seen in this figure, diffuse transmittance increases as μ_a and/or μ_s^* decreases, and total reflectance increases as μ_a decreases and/or μ_s^* increases.

The collimated transmittance T_c was also calculated by the Monte Carlo code, and the corresponding ratio $B = T_c/T$ was found.

In figure 4 the calculated ratios $B = T_c/T$ versus g, for various μ_a and μ_s^* values, are shown, Also the plots fitted to the data are presented. As expected B increases with increasing value of g, since the higher g is, the smaller the light scattering solid angle is. When $g = 1$, all the scattered photons follow the forward direction, and so the total transmittance is equal to the collimated one.

3.1. Indirect determination of optical microscopic coefficients

The results presented in figure 3 can be used to determine optical microscopic coefficients for a tissue type, using experimental data of the macroscopic parameters of transmittance and total reflectance for that tissue type. Since no analytical solution exists for equations (1) and (2) graphical solutions were employed. The procedure proposed is in effect a graphical inversion of the relations (1) and (2).

To simplify the procedure, isotransmittance and isoreflectance curves were drawn in figure 5(a) and (b) from the fitted curves in figures 3(a) and (b) respectively. From these curves, a value of diffuse transmittance T and/or total reflectance R can be associated with an infinite number of pairs of values of μ_a and μ_s^*.

The combination of the isotransmittance and isoreflectance curves is presented in figure 6. From these graphs, it appears that if the values of T and R are known, a unique pair of μ_a and μ_s^* can be obtained, as the coordinates at the intersection of the corresponding isotransmittance and isoreflectance curves.

The anisotropy factor g can also be determined, if the collimated transmittance T_c is measured, and consequently the ratio $B = T_c/T$ is known. Then, from figure 4, g is found as the abscissa of the intersection of the horizontal line corresponding to that value of B and the curve corresponding to the values of μ_a and $\mu_s^{*\prime}$ which were found from figure 6.

The graphs of figures 4(a), 4(b) and 6 can only be used in the case where the macroscopic quantities will be measured under the same experimental conditions as used as input for the Monte Carlo calculations, namely for 1 cm sample thickness, 1 cm beam diameter and in the case of collimated transmittance measurements 1 cm beam cross-sectional area at the exit close to the tissue sample. At least for liquid samples and certain

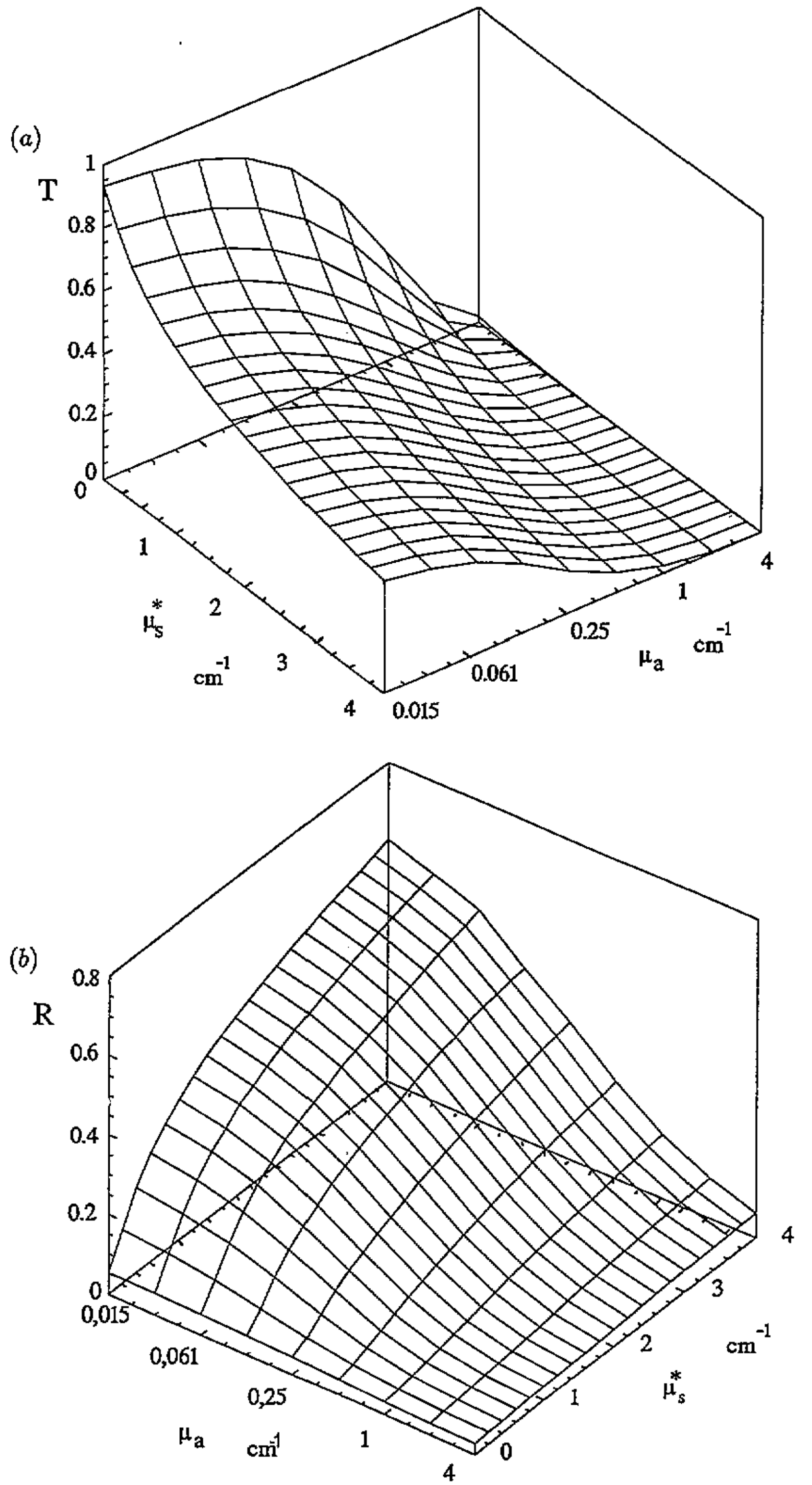

Figure 3. Three-dimensional representations of (*a*) diffuse transmittance T and (*b*) total reflectance R as functions of μ_s^* and μ_a. The slab thickness is 1.0 cm.

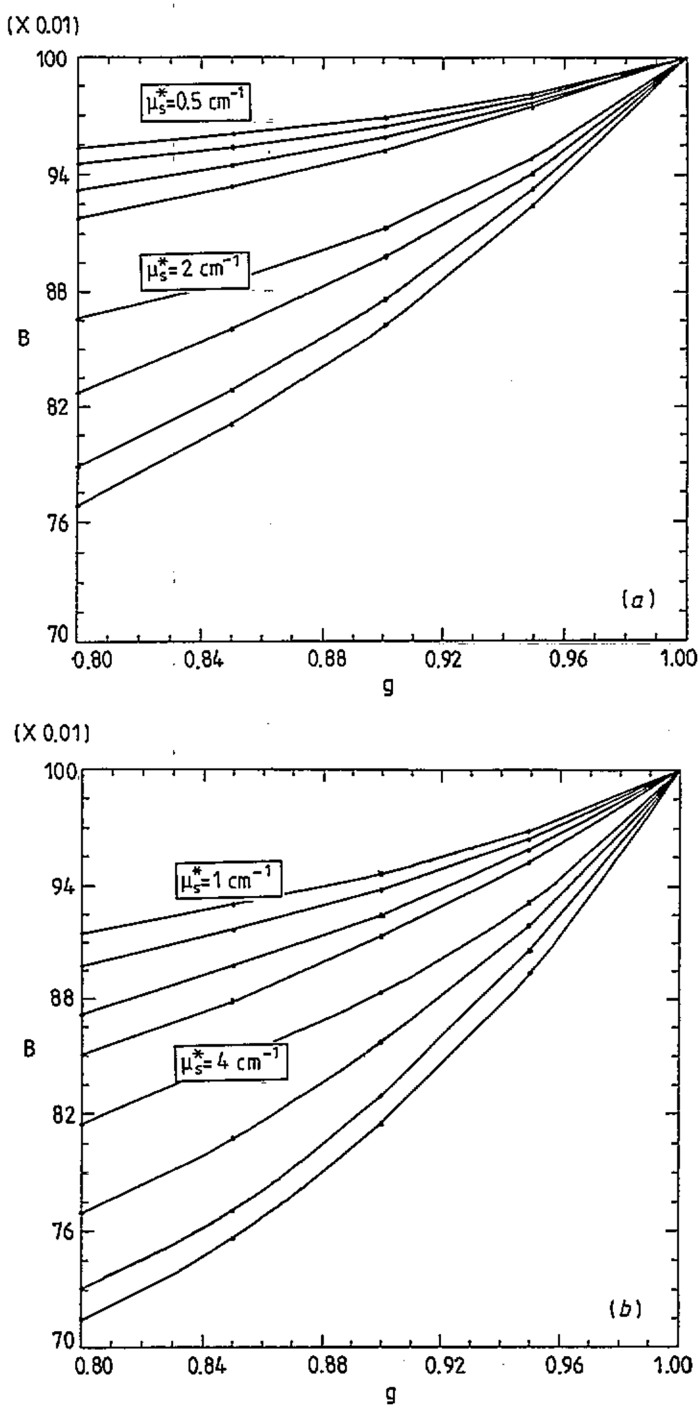

Figure 4. B ratio as a function of g with respect to μ_s^* and μ_a. Four families of curves are presented. (*a*) B for $\mu_s^* = 0.5$ cm^{-1} (upper family) and $\mu_s^* = 2$ cm^{-1} (lower family). (*b*) B for $\mu_s^* = 1$ cm^{-1} (upper family) and $\mu_s^* = 4$ cm^{-1} (lower family). All four families have been calculated for the same set of values for μ_a, that is (from top to bottom) 0.015 cm^{-1}, 0.061 cm^{-1}, 0.25 cm^{-1} and 1 cm^{-1}. The slab thickness is 1.0 cm.

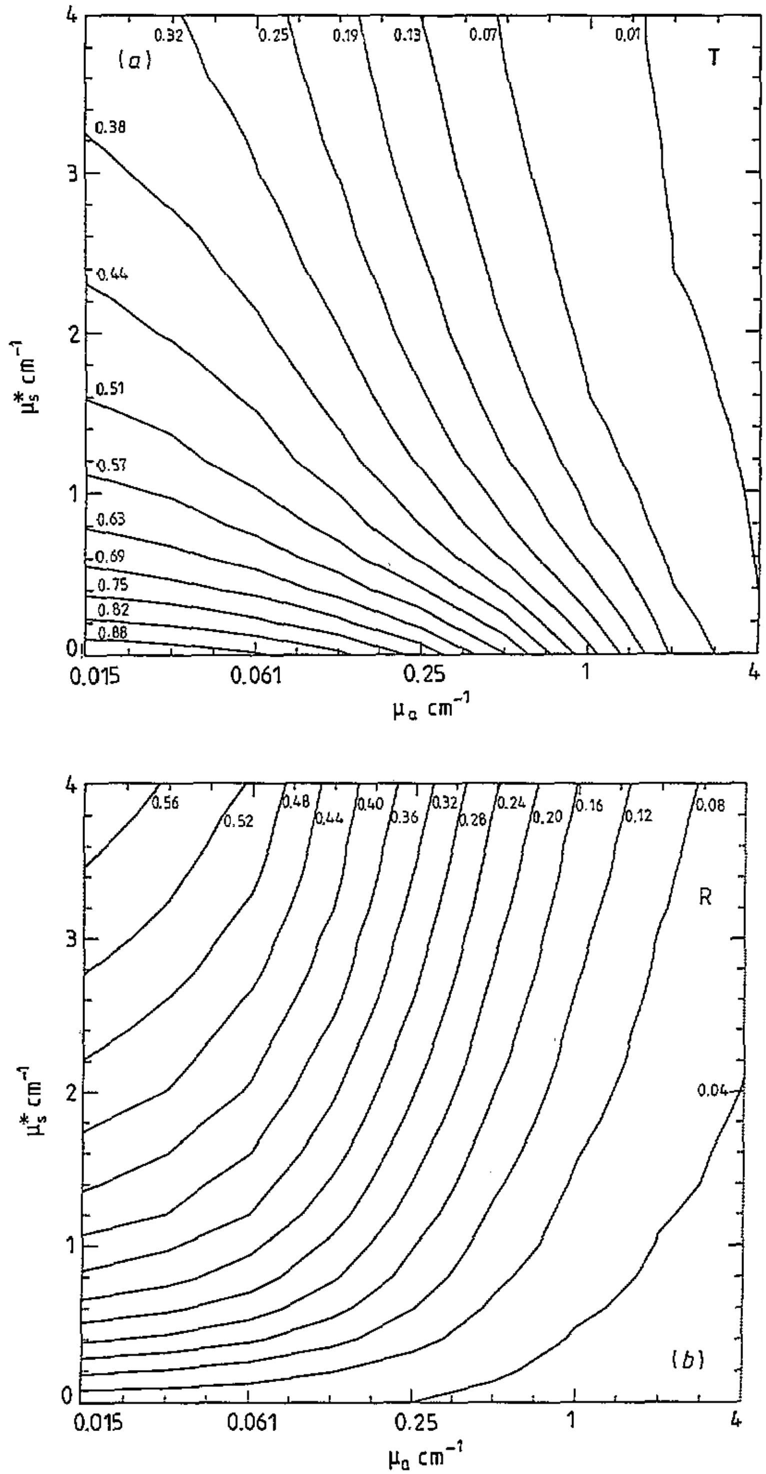

Figure 5. (*a*) Isotransmittance and (*b*) isoreflectance curves derived from figure 3(*a*) and (*b*) respectively. Each value of diffuse transmittance, T, and total reflectance, R, corresponds to an infinite number of μ_s^* and μ_a pairs. The slab thickness is 1.0 cm.

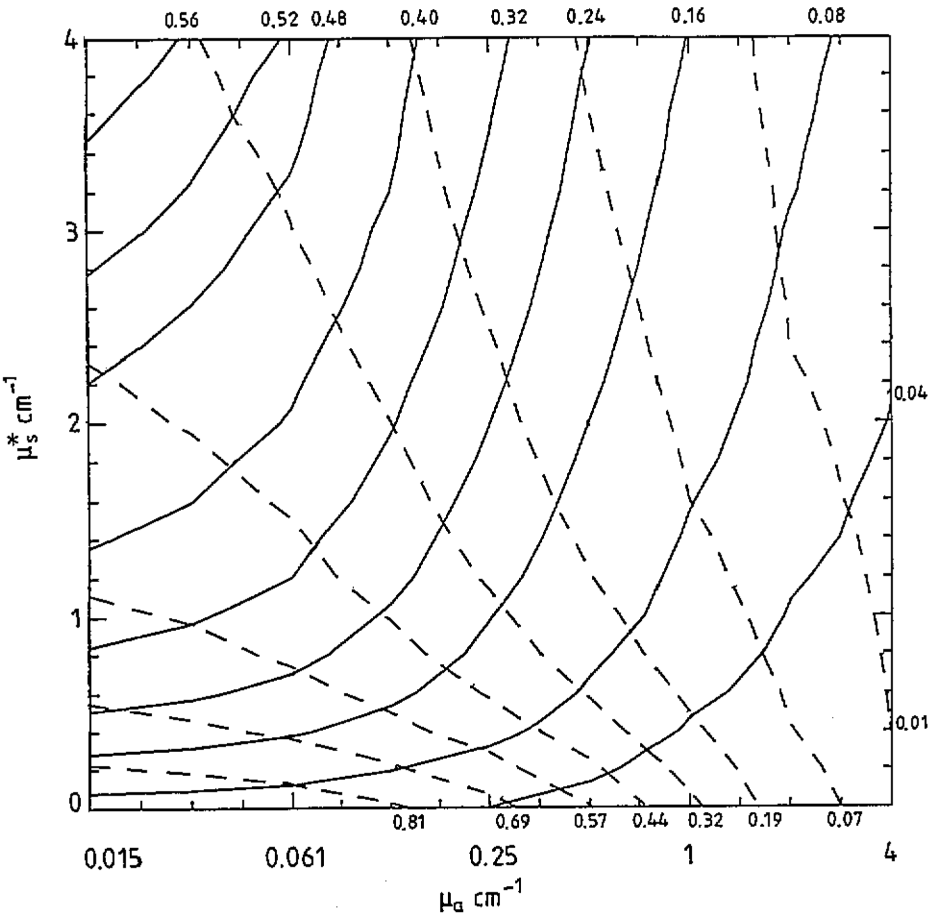

Figure 6. Superimposed isotransmittance (dashed lines) and isoreflectance (continuous lines) curves from figure 5. Only one pair of μ_s^* and μ_a values corresponds to any given pair of T and R, and it is read out from the intersection of the superimposed graphs. The slab thickness is 1.0 cm.

other situations, these conditions can easily be realized by experimenters. In the case of different experimental conditions new graphs have to be obtained.

All the fitting parameters to the equations (1) and (2) were obtained on the assumption that the slab thickness was 1.0 cm.

3.2. The validity of the results and the accuracy of the code

The validity of the results must be verified by comparison with measured values that have been published. For such comparisons one needs to know the experimental conditions, such as slab thickness and beam diameter, as well as the measured macroscopic quantities, such as diffuse transmittance T and total reflectance R. Such complete sets of data have been published by some investigators (see e.g. Peters *et al* 1990, Derbyshire *et al* 1990, Splinter *et al* 1989, 1991).

The values of diffuse transmittance and diffuse reflectance measured by Peters *et al* (1990) and Derbyshire *et al* (1990) have been used in order to check the validity of the present method. The corresponding microscopic parameters as calculated by this method, simulating their experimental conditions, were compared to those found by the above investigators. This comparison is presented in table 3. Table 3 shows that there is an excellent agreement of the results obtained by this method and the results of both Peters *et al* (1990) and Derbyshire *et al* (1990). No comparison was attempted with the results of Splinter *et al* (1989, 1991), since these investigators are using a diffuse source instead of a collimated one.

Table 3. The validity of the results.

Peters *et al* (1990)			
		This work	Peters *et al*
Reflectance and transmittance measurements			
Sample thickness 1.0 mm			
Beam diameter 3.0 mm			
Detector aperture 12.7 mm	μ_a (mm^{-1})	0.063	0.062 ± 0.005
Wavelength 900 nm			
Measured values: $T = 0.42$; $R = 0.24$	μ_s^* (mm^{-1})	1.02	0.99 ± 0.20
Total attenuation measurements			
Sample thickness 0.15 mm			
Beam diameter 4.0 mm			
Detector aperture 0.2 mm	g	0.97	≈ 0.964[a]
Wavelength 900 nm			
Measured value: $T_c \approx 0.406$[a]	μ_s (mm^{-1})	34	≈ 30[a]
Derbyshire *et al* (1990)			
		This work	Derbyshire *et al*
Sample thickness 1.4 mm			
Beam diameter 1.0 mm			
Detector aperture 20 mm	μ_a (mm^{-1})	0.0504	0.052
Wavelength 1.06 μm			
Measured values: $T = 0.655$; $R = 0.213$	μ_s^* (mm^{-1})	0.438	0.448

[a] Values taken from graphs.

The accuracy of the code has been checked by comparing its results to published values of macroscopic properties, which were calculated from microscopic properties. This comparison is summarized in table 4. Calculations based on diffusion theory (Patterson *et al* 1989), or derived by Monte Carlo (Key *et al* 1991), as well as experimental results (Yoon *et al* 1987) have been included in this comparison.

From table 4, it is clear that the results published are in very good agreement with the results obtained by this Monte Carlo code.

Finally, the values of specular reflectance f, calculated by the Monte Carlo code, were between 0.027 and 0.028, for any combination of μ_s, μ_a and g. This is in very good agreement with the theoretical predicted value of 0.0278, calculated by the formula $f = (1-n)^2/(1+n)^2$ (Wilson and Jacques 1990) when $n = 1.4$.

4. Discussion

In order to carry out treatment planning in PDT (Wilson and Patterson 1986) certain quantities, such as the space irradiance distribution, the energy fluence rate and the effective

Table 4. The accuracy of the code.

Patterson *et al* (1989)

$N' = \mu_s(1-g)/\mu_s$	Diffuse reflectance: Diffusion theory	Diffuse reflectance: This work	Boundaries
40	0.5	0.50	
240	0.75	0.74	mismatched
1667	0.87	0.87	
40	0.65	0.67	
240	0.82	0.83	matched
1667	0.90	0.92	

Key *et al* (1991)

Slab thickness (mm)	μ_a (cm^{-1})	μ_s^* (cm^{-1})	ln (*T*): Calculated values	ln (*T*): This work
25	0.1	7.5	−4.4	−4.414
10	0.5	7.5	−4.9	−5.035

Yoon *et al* (1987)

Slab thickness (mm)	μ_a (cm^{-1})	μ_s (cm^{-1})	*g*	ln (*T*): Experimental value	ln (*T*): This work
0.4	0.55	315.45	0.882	−0.60	−0.604
1.0	0.55	315.45	0.882	−1.0	−1.055
1.6	0.55	315.45	0.882	−1.5	−1.824

absorbed dose rate (Profio and Doiron 1987) and its distribution within the tissues, must be used. When optical tissue microscopic properties are known, the above quantities can be determined, by applying an adequate theory or model of light propagation in tissues.

As was mentioned in the previous section, the present method produces results that are in agreement with those of other investigators, when their experimental conditions are used. However, it must be pointed out that broad ranges in optical properties for each tissue type have been reported. This diversity has also been pointed out by other investigators (Splinter *et al* 1991, Cheong *et al* 1990, Wilson *et al* 1987). In table 5 the optical properties for three tissue types are presented as measured by different investigators. Such discrepancies in optical coefficients, which are frequent in the literature, indicate the sensitivity and vulnerability of such measurements to variations in samples, experimental set-up, detection apparatus, boundary conditions and the light propagation model used.

By the present method the microscopic properties can be derived from measurements of the macroscopic parameters utilizing the results of the Monte Carlo code. Thus the method described in this work provides graphical 'inverse' solutions of the relations governing the light propagation in tissues. The need for such solutions has been pointed out in the literature (Flock *et al* 1989, Wang and Jacques 1993, Wilson *et al* 1987).

The present method can be applied in both direct or indirect techniques. It does not depend on limitations introduced by assumptions and approximations when using theoretical models. It also can be applied for any tissue type, detector geometry and experimental apparatus. Finally, the accuracy of the method is very good over a wide, unlimited in practice, range of values of optical properties.

Table 5. Measured tissue optical properties.

Tissue	λ (nm)	μ_a (cm^{-1})	μ_s (cm^{-1})	g	μ_t (cm^{-1})	μ_{eff} (cm^{-1})	Reference
Human liver	630	3.2	414	0.95	417.2	86.0	Andreola *et al* 1989
Human liver	630					11.0	Eichler *et al* 1977
Human dermis	630	1.8	241.2		243		Andreola *et al* 1989
Human dermis	630	5	60		65		Anderson and Parrish 1981
Human dermis	630	1–4	17		18		van Gemert *et al* 1989
Muscle bovine	633	0.4	7.9	0.3	8.3	2.7	Marijnissen and Star 1987
Muscle bovine	633	1.5	119	0.94	121	6.2	Wilson and Patterson 1986

For the above reasons the method proposed in this work can be used as a reference when optical parameter measurements are performed, in order to acquire values of microscopic properties for use in light dosimetry.

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